

Infinite Series Solutions

Analytic function

A function $f(x)$ is **analytic** at x_0 if its Taylor series about

$x_0, \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)(x - x_0)^n}{n!}$ converges to $f(x)$ in some

neighborhood of x_0 .

Ordinary Point for the differential equation

Consider the differential equation

$$b_2(x)y'' + b_1(x)y' + b_0(x)y = 0$$

If we divide by $b_2(x)$

$$y'' + \frac{b_1(x)}{b_2(x)}y' + \frac{b_0(x)}{b_2(x)}y = 0$$

The standard form

$$y'' + P(x)y' + Q(x)y = 0$$

$$P(x) = \frac{b_1(x)}{b_2(x)}$$

$$Q(x) = \frac{b_0(x)}{b_2(x)}$$

A point x_0 is **an ordinary point** if both $P(x)$ and $Q(x)$ are analytic at x_0 . If either $P(x)$ or $Q(x)$ is not analytic at x_0 then x_0 is a **singular point**.

Regular Singular Point

Consider the second order linear differential equation in standard form

$$P(x)y'' + Q(x)y' + R(x)y = 0 \quad (1)$$

$x = 0$ is a regular singular point of Eq (1) means that

$$x \frac{Q(x)}{P(x)} \quad \text{and} \quad x^2 \frac{R(x)}{P(x)} \quad \text{have finite limit as } x \rightarrow 0.$$

and are analytic at $x = 0$.

Singular points which are not regular are called irregular.

Example 1

Determine whether $P(x) = \frac{-x}{(1-x^2)}$ and $Q(x) = \frac{1}{(1-x^2)}$ are analytic at $x = 0$.

We need to determine whether the Maclaurin series of these functions converge in some interval around $x = 0$.

$$P(x) = \frac{-x}{(1-x^2)} = -x - x^3 - x^5 - \dots$$

$$Q(x) = \frac{1}{(1-x^2)} = 1 + x^2 + x^4 + \dots$$

These expansions converge when $|x^2| < 1$.

Therefore both functions are analytic at $x = 0$.

Example 2

Consider the differential equation

$$2x^2 y'' - xy' + (1+x)y = 0$$

We can show that $x = 0$ is a regular singular point.

$$y'' - \frac{x}{2x^2} y' + \frac{(1+x)}{2x^2} y = 0$$

$$y'' - \frac{1}{2x} y' + \frac{(1+x)}{2x^2} y = 0$$

$$P(x) = -\frac{1}{2x}$$

$$Q(x) = \frac{(1+x)}{2x^2}$$

Neither of these functions are analytic at $x = 0$.

$$xP(x) = -\frac{1}{2}$$

are finite at $x = 0$.

$$x^2Q(x) = \frac{(1+x)}{2}$$

Therefore $x = 0$ is a regular singular point.

Example 3

Consider the differential equation

$$8x^2 y'' + 10xy' + (x-1)y = 0$$

We can show that $x = 0$ is a regular singular point.

$$y'' + \frac{10x}{8x^2} y' + \frac{(x-1)}{8x^2} y = 0$$

$$y'' + \frac{5}{4x} y' + \frac{(x-1)}{8x} y = 0$$

$$P(x) = \frac{5}{4x}$$

$$Q(x) = \frac{(x-1)}{8x^2} = \frac{1}{8x} - \frac{1}{8x^2}$$

Neither of these functions are analytic at $x = 0$.

But

$$xP(x) = \frac{5}{4}$$

are finite at $x = 0$.

$$x^2Q(x) = \frac{(x-1)}{8}$$

Therefore $x = 0$ is a regular singular point.